SecondChance

Competing-risk survival forecasting of shelter adoption, with calibration and drift monitoring

Davis Higgins · DTSC 3601, UNC Charlotte · 29 September 2026

136,898 shelter episodes · nine years of Austin Animal Center records · 20,165 held out



Shelters run out of attention, not kennels

Photography, foster recruitment, behaviour support and outreach are allocated by whoever happens to notice.

16,000

animals a year through one shelter,
the largest no-kill municipal shelter
in the United States

The animals nobody happens to notice are the ones who wait months.

1. **Will this animal go home this week?**

   P(adopted in 7, 14, 30 days)

2. **Will it still be here in two months?**

   P(long stay past 60 days)

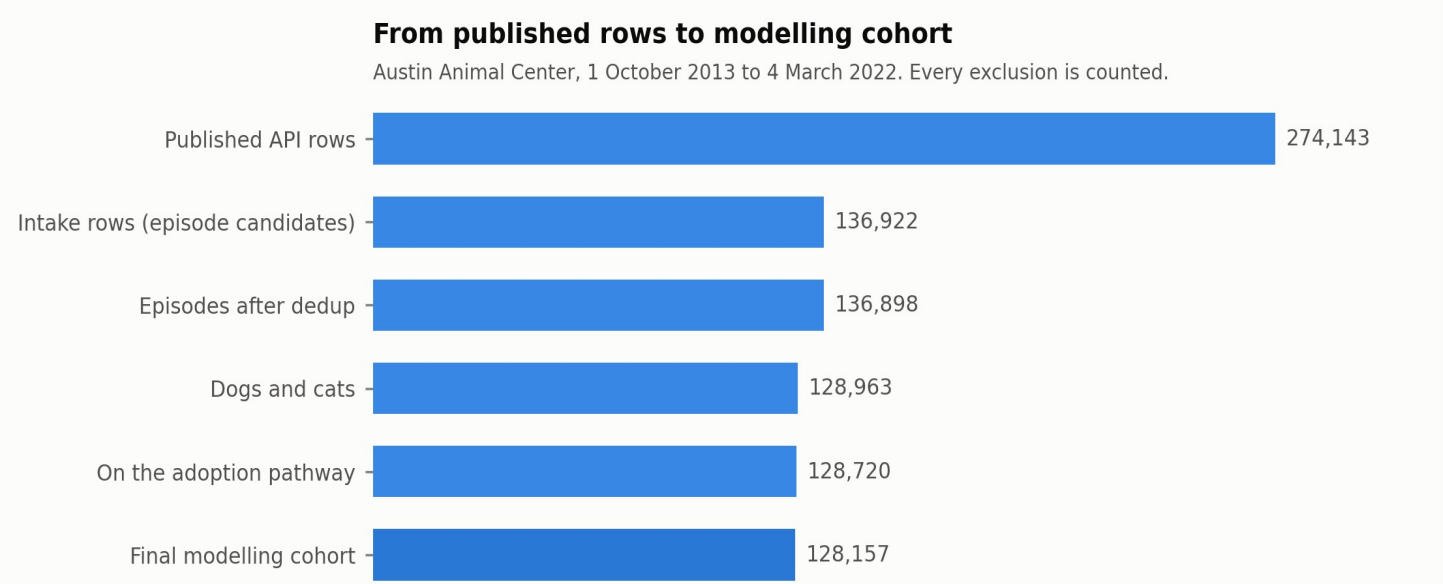
3. **How long, exactly?**

   Full cumulative incidence curve

Answered at the intake desk, from 56 features, none of which require anyone to watch the animal.

Two event logs, no episode key, and repeat visitors

A join on animal ID alone produces a cross product whose durations look entirely plausible.



The audit

Episodes built	**136,898**
Closed by an outcome	**135,669 (99.1%)**
Matches rejected	**42**
Negative durations	**0**
Species mismatches	**0**
Suspected lost paperwork	**622**

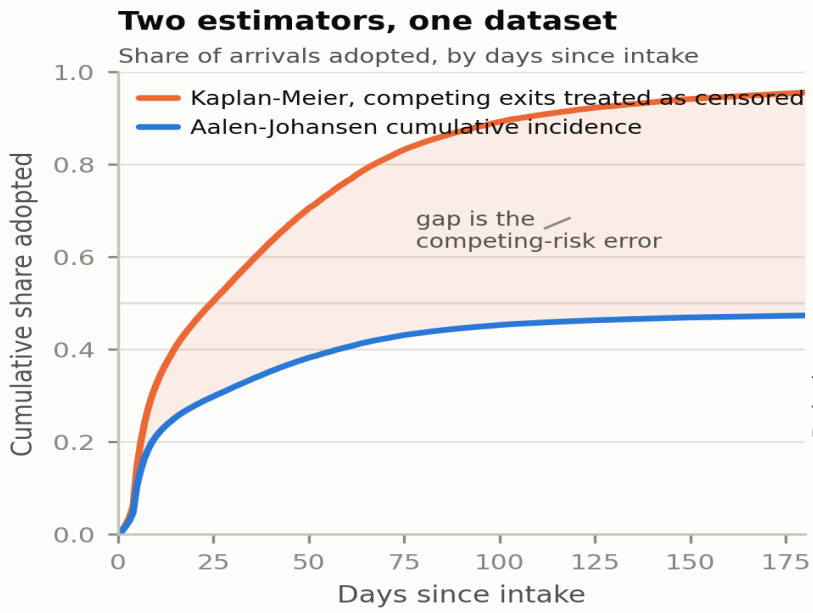
Shuffled-label control:

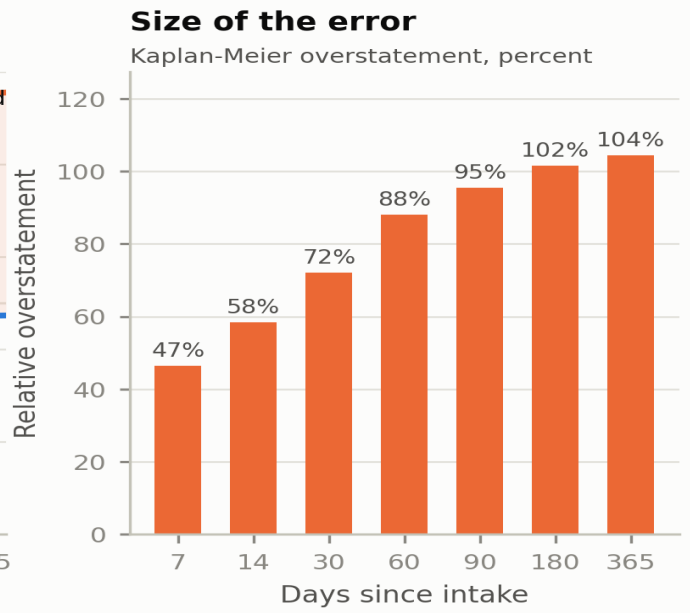
AUC 0.5023

no feature encodes the outcome

The estimator the field uses is wrong by 72 percent

Half of all exits are not adoptions. Treating a transfer as censoring assumes the animal is still waiting to be adopted.





At one year

98.8%

what Kaplan-Meier says

48.3%

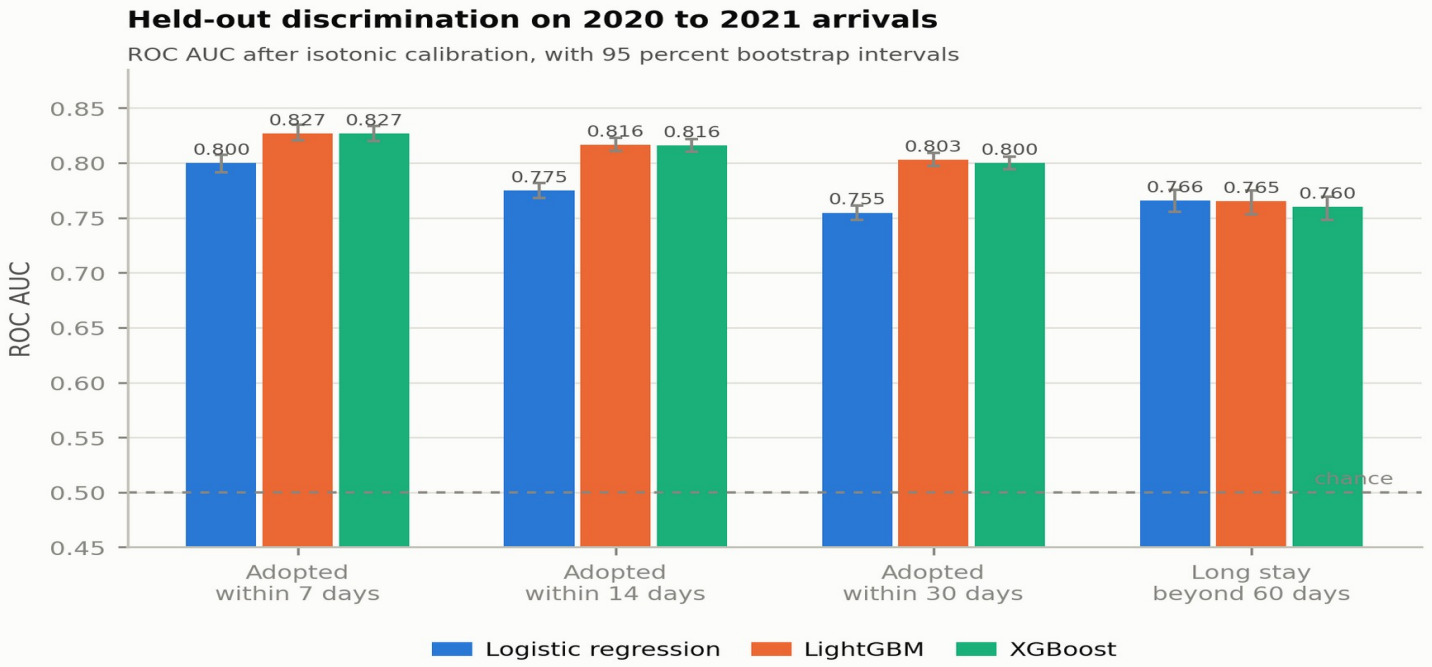
what actually happens

Fewer than half of all arrivals are ever adopted from this building.

Same error per prediction: our survival model without the correction predicts a mean 30-day adoption probability of 0.638. With it, 0.403. Observed: 0.372.

Calibrated, and tested on animals the model never saw

Train through 2018, calibrate on 2019, test on 2020 to 2022. The split spans a known operational discontinuity on purpose.



0.803
held-out AUC,
30-day adoption

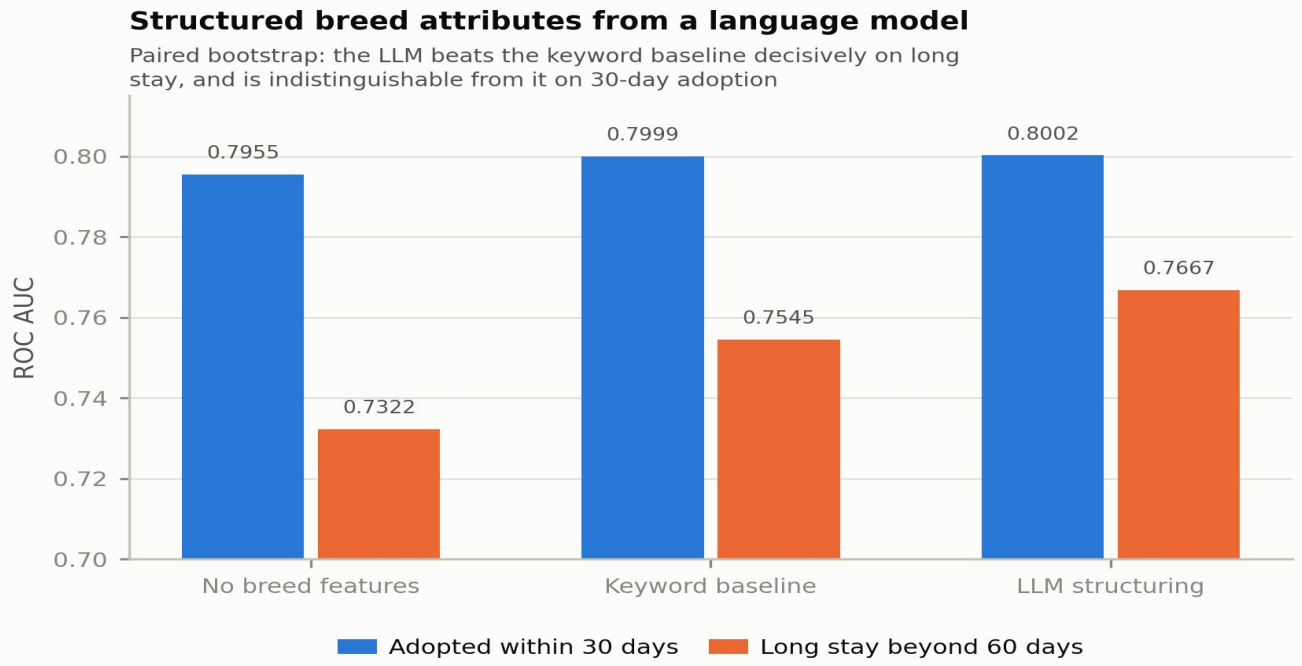
0.023
expected calibration error
(the score means what it says)

4.2×
better than chance in the
top 50 of the long-stay queue

48 percent of the model's top 50 really did stay past 60 days, against a base rate of 11.6 percent.

The language model earned its place on one target

2,736 free-text breed strings collapse to 254 atomic tokens. Claude labels each once with eleven structured attributes.



Long stay past 60 days

+0.012 AUC over a keyword baseline

p < 0.001, paired bootstrap. +0.035 over dropping breed entirely.

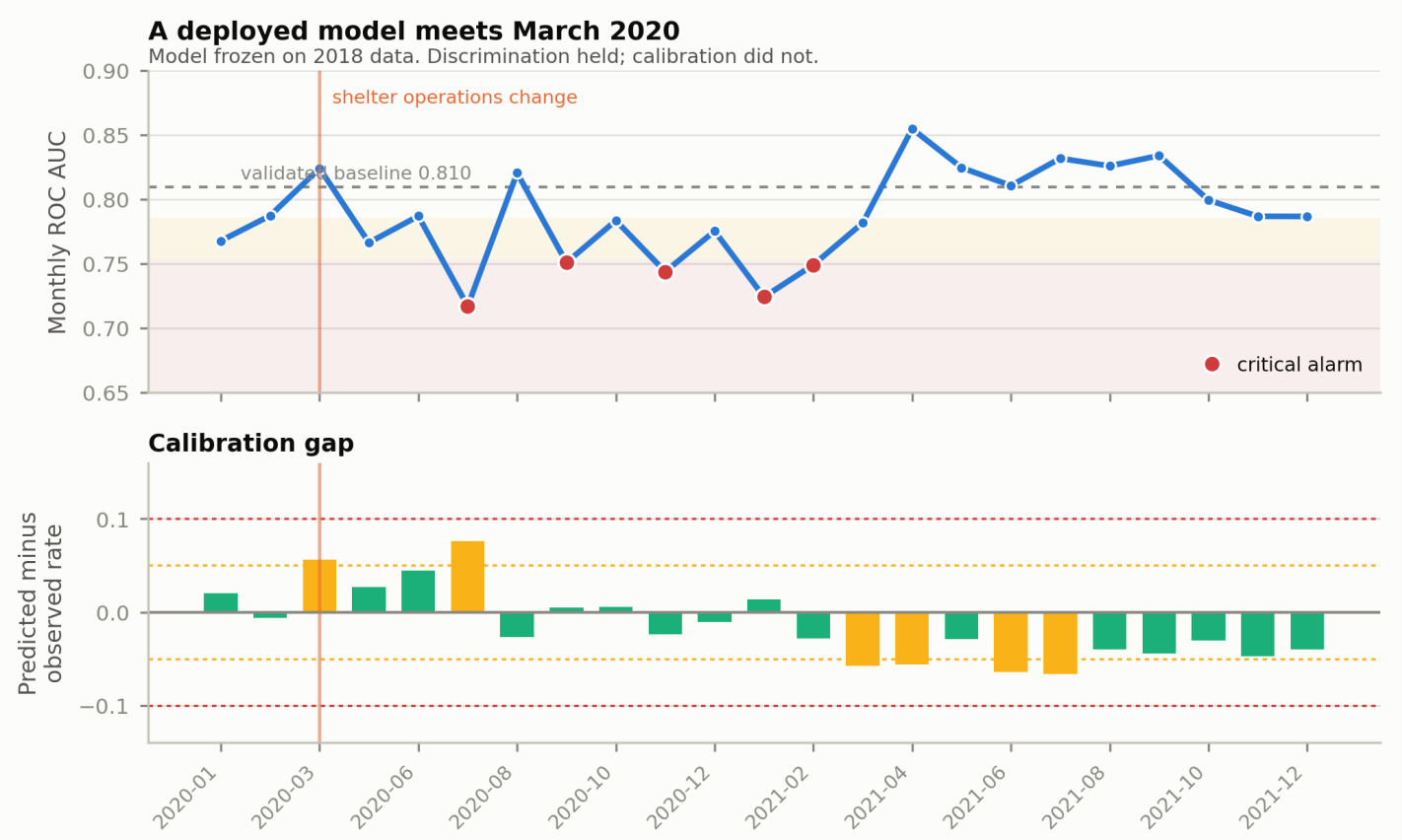
Adoption within 30 days

+0.0003 AUC. Nothing.

p = 0.73. A regular expression does the job for this question.

Breed predicts who is still here in two months far better than it predicts a fast adoption. Weather, our fourth data source, did not help either target and we say so.

March 2020: the ranking held, the probabilities did not



Discrimination

0.778 → 0.789

mean monthly AUC, before and after. It did not degrade.

Calibration

7.6 points off

in July 2020. It broke, and the ranking metric never noticed.

Input drift fired a quarter earlier than performance drift could.
The 30-day label does not exist for 30 days, so outcome monitoring is structurally six weeks behind its own cause.

A lifecycle, not a notebook

Serverless end to end, with the monitoring loop and the human gate that make it a system.

1 Ingest

Four public APIs on a schedule into a versioned, encrypted S3 landing zone. Raw responses kept forever.

2 Warehouse

Glue and Redshift Serverless at 8 RPU. Episode reconstruction, occupancy, and the analytics grain.

3 Label

Bedrock structures any breed token the reference has not seen. Prompt versioned, output committed.

4 Train

SageMaker transient jobs. Every model registered with its data cutoff, metrics and feature hash.

5 Serve

API Gateway and Lambda. Five routes. Every response carries the model version, data freshness and use policy.

6 Watch

Weekly job: 19 monitored features, rolling AUC, calibration gap. Six CloudWatch alarms into email.

Measured cost: $34.07 a month with no persistent endpoint. A real-time endpoint would add $169.

Direction is an engineering decision

What we found

Dogs carrying a bully-type breed label reach 25.1 percent 30-day adoption against 34.0 percent for others. The model is also worse for them (AUC 0.754 against 0.807) and under-promises their chances by 4.8 points.

Olson et al. measured 50 percent sensitivity for that label against DNA. This is a finding about the label, not the animals.

What we refused to build

The same machinery that ranks animals for extra help could rank them for less. The 2002 paper that founded this literature framed the problem as adoption versus euthanasia.

So the use policy ships in every API response, the interface has no ranking that surfaces animals for removal, and the public page shows no probability at all.

A low predicted adoption probability is a request for more help, not less.

Black coat colour, incidentally, did not slow adoption at all: 34.1 percent against 32.3. A model that encoded the folk belief would have begun making it true.

Four things I would defend, and three I would not

Findings

1. The standard estimator overstates 30-day adoption by 72 percent on this data.
2. Calibration and discrimination fail separately. Calibration failed first.
3. The language model earns its keep on one target and not the other.
4. Three whole classes of feature will silently break a drift detector.

Limitations

- No behavioural data, which is where the remaining signal is.
- Breed is a staff guess with 50 percent measured sensitivity.
- We can show the queue works. We cannot show that acting on it changes outcomes.

Next: a photograph-quality signal, a second shelter, and a prospective evaluation.